Exam

MNF130 Discrete Structures

Faculty of Mathematics and Natural Sciences
University of Bergen

20 May 2022, 15:00–18:00

- For each solution make sure to **clearly state which problem you are solving**.
- **No aids** (formula sheets, calculator, phone, ...) are permitted during the exam.

- As most of the exams will be graded by people not knowing Norwegian, you are strongly encouraged to **write** your answers **in English**. You are allowed to write in Norwegian or another Scandinavian language.

- The exam has **8 questions** on 6 pages, and is scored on 60 points.

Good luck!

1 (8 points)

a) Fill out the truth table:

p	q	r	$p \to q$	$q \to r$	$(p \to q) \to r$	$p \to (q \to r)$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

b) Let p, q, r be propositions. Are the compound propositions $(p \to q) \to r$ and $p \to (q \to r)$ logically equivalent? Explain why/why not.

c) Let p, q be propositions. Use basic logical equivalences to prove that the compound propositions $(p \wedge q) \to p$ is a tautology.

d) Let $P(m, n)$ be "$n + m \neq 0$" where the domain (universe of discourse) is the set of integers. What are the truth values of $\exists n \forall m P(n, m)$ and $\forall n \exists m P(n, m)$? Explain your answer.

2 (8 points)

a) Let A, B, C be sets. Prove using set identities that

$$\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}.$$

b) Let $f(n) = |n|$ be a function from the set of non-negative integers to the set of non-negative integers. Is f one-to-one (injective)? Onto (surjective)? A one-to-one correspondence (bijective)? Explain why/why not.

c) Verify that $a_n = 2 \cdot 3^n$ is a solution to the recurrence relation $a_0 = 2$, $a_1 = 6$ and

$$a_n + a_{n-1} - 12a_{n-2} = 0,$$

for $n \geq 2$.

3 (8 points)

a) Let a, b, c, m, n be integers. Let $a|b$ and $a|c$. Show that

$$a|(mb+nc).$$

b) Use the extended Euclidean algorithm to find integers u and v such that

$$\gcd(240,46) = u \cdot 240 + v \cdot 46.$$

c) Consider designing a card-based boardgame. The game should be possible to play for 2, 3, 4, 5 and 6 players respectively. No matter the number of players, the number of cards should be divisible by the number of players. What is the minimum number of cards that fulfills this requirement?

4 (8 points)

Consider the function f with non-integers as the domain, defined by $f(0) = 1$, $f(1) = 1$ and

$$f(n) = \sum_{k=0}^{n-1} f(k),$$

for $n \geq 2$.

a) Compute $f(n)$ for a few values and make a hypothesis for a formula for $f(n)$, for $n \geq 2$.

b) Prove you formula from $a)$ using induction. Make sure to clearly divide up the proof into the base case, the induction hypothesis and the induction step.

5 (8 points)

a) Prove using either algebra or combinatorial reasoning that

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1},$$

where n, k are positive integers and $n \geq k$.

b) In a group with n people, every person shakes hand with a subset of the other people. Show that at least two persons shake hands with the same number of people.

c) The complete graph K_n, $n \geq 1$ is a simple, undirected graph with n nodes, in which there is an edge between every pair of nodes. How many edges are there in K_n?

6 (8 points)

a) Let R be a binary relation on the set of ordered pairs of positive integers defined by

$$((a,b),(c,d)) \in R \leftrightarrow ad = bc.$$

Show that R is an equivalence relation.

b) Consider a relation R_2 on a set of 3 elements, defined by the matrix

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Determine whether R_2 is reflexive, symmetric and antisymmetric respectively.

7 (8 points)

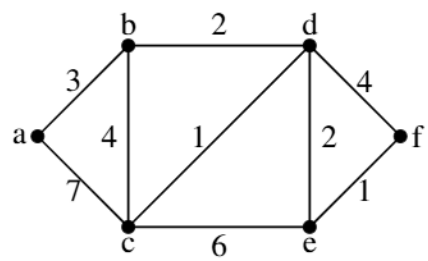
Consider the list $[4,2,7,1,3,5,8,6]$. Sort the list by

a) applying the merge sort algorithm.

b) by building a binary search tree and then traversing the binary search tree in the correct order.

8 (4 points)

For the weighted graph shown, rank the vertices by their shortest-path length from *a* and give a shortest path from *a* to each vertex.



Vertex	Shortest length	Shortest path